

Internship Report — Day 10

Intern Name: Syed Fuzail

Internship Role: Cloud Engineer Trainee Intern

Organization: Spektra Systems

Day: 10

Task Title: Deployment of Secure Reverse Proxy Architecture using Azure ARM Template

1. Objective of the Task

The objective of today's task was to design and deploy a **secure multi-tier cloud architecture** using an **Azure ARM (Azure Resource Manager) template**. The deployment involved automating infrastructure provisioning through Infrastructure as Code (IaC) principles and understanding dependency management between Azure resources.

The architecture aimed to implement a **Reverse Proxy setup** where:

- A public-facing virtual machine acts as an **Nginx Proxy Server**
 - A backend application virtual machine remains **hidden in a private subnet**
 - Network security controls restrict direct public access to internal resources
-

2. Technologies and Services Used

- Microsoft Azure ARM Templates
- Azure Virtual Machines (Linux – Ubuntu 22.04)
- Azure Virtual Network (VNet)
- Subnets (Public and Private)
- Network Security Groups (NSG)
- Public IP Address
- Azure VM Extensions (Custom Script Extension)

- Nginx Web Server
-

3. Architecture Overview

The deployed architecture consisted of:

Network Layer

- A Virtual Network named **LinuxProxyVNet**
- Two subnets:
 - **Proxy-Subnet (Public Subnet)** – Hosts the proxy VM
 - **App-Subnet (Private Subnet)** – Hosts the internal application VM

Security Layer

- Network Security Group allowing:
 - SSH (Port 22)
 - HTTP (Port 80)
- Backend VM had **no public IP**, ensuring isolation.

Compute Layer

Two Linux Virtual Machines were deployed:

1. **Nginx Proxy VM**
 - Publicly accessible
 - Receives external traffic
 - Forwards requests internally
2. **Hidden App VM**
 - Private IP only (10.0.2.4)

- Hosts internal web application
 - Accessible only via proxy server
-

4. Implementation Process

Step 1: ARM Template Creation

A JSON ARM template was created containing:

- Parameters for username, password, VM size, and location
- Variables defining resource names and network configuration
- Resources defining networking and compute infrastructure

Step 2: Dependency Configuration

The `dependsOn` property was implemented to ensure correct deployment order:

- NSG created before Virtual Network
- VNet created before Network Interfaces
- NICs created before Virtual Machines
- Application VM configured before Proxy VM setup

This ensured Azure deployed resources sequentially without failures.

Step 3: Networking Deployment

- NSG rules were applied to both subnets.
- Public IP assigned only to proxy VM.
- Static private IP assigned to application VM.

Step 4: Virtual Machine Deployment

Ubuntu Linux VMs were automatically provisioned using ARM template image references.

Step 5: Automated Server Configuration

Azure Custom Script Extensions executed commands automatically:

Application VM Configuration

- Installed Nginx

Created a web page:

Secret App Server - Hidden in Private Subnet

-

Proxy VM Configuration

- Installed Nginx
- Configured reverse proxy
- Forwarded traffic to private server (10.0.2.4)

5. Deployment Execution

The deployment was performed using PowerShell:

```
New-AzResourceGroupDeployment `
  -ResourceGroupName "syedfuzail2009-rg" `
  -TemplateFile ".\vvvmmmm.json" `
  -Verbose
```

The verbose mode allowed observation of:

- Resource creation sequence
 - Dependency execution
 - Provisioning status
-

6. Key Observations

- Azure deploys resources based on dependency mapping rather than template order.
 - Proper `dependsOn` configuration prevents deployment conflicts.
 - Private subnet architecture improves security by eliminating direct internet exposure.
 - VM Extensions enable automatic configuration without manual SSH login.
 - Reverse proxy architecture acts as a secure gateway between internet users and backend services.
-

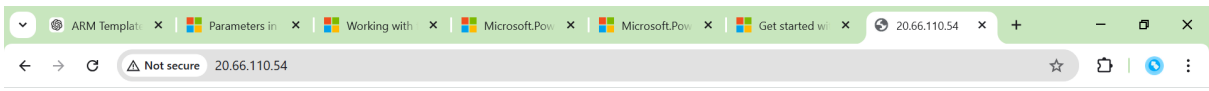
7. Learning Outcomes

Through this task, I learned:

- Practical implementation of **Infrastructure as Code (IaC)** using ARM templates.
 - Importance of resource dependency management.
 - Designing secure cloud network architectures.
 - Reverse proxy implementation using Nginx.
 - Automating server configuration using Azure VM Extensions.
 - Secure separation of public and private workloads in Azure networking.
-

8. Conclusion

Day 10 focused on deploying a production-style secure architecture using ARM templates. The task demonstrated how Azure automation can provision complete infrastructure, configure servers, and enforce security controls without manual intervention. The reverse proxy model provided practical exposure to real-world cloud security and networking concepts commonly used in enterprise environments



Secret App Server - Hidden in Private Subnet

Microsoft Azure portal screenshot showing the 'syedfuzail2009-rg' resource group. The 'Resources' tab is selected, displaying a list of resources in the 'West US' location.

Name	Type	Location
Hidden-App-VM	Virtual machine	West US
Hidden-App-VM_disk1_6a649b84102146a8f949c3da88163	Disk	West US
HubSpokeVNet	Virtual network	West US
Linux-NSG	Network security group	West US
LinuxProxyVNet	Virtual network	West US
Ngini-Proxy-VM	Virtual machine	West US
Ngini-Proxy-VM_disk1_99ffdf2cbb66423c981d34ca21566	Disk	West US
nic-app	Network Interface	West US
nic-proxy	Network Interface	West US
Proxy-Public-IP	Public IP address	West US

